



CLOUD ADOPTION PLAYBOOK

ACCELERATOR PLATFORM WITH TURNKEY SOLUTIONS

*ACCELERATING THE ENTERPRISE CLOUD ADOPTION AND DIGITAL MODERNIZATION JOURNEY
SECURELY AND EFFICIENTLY*

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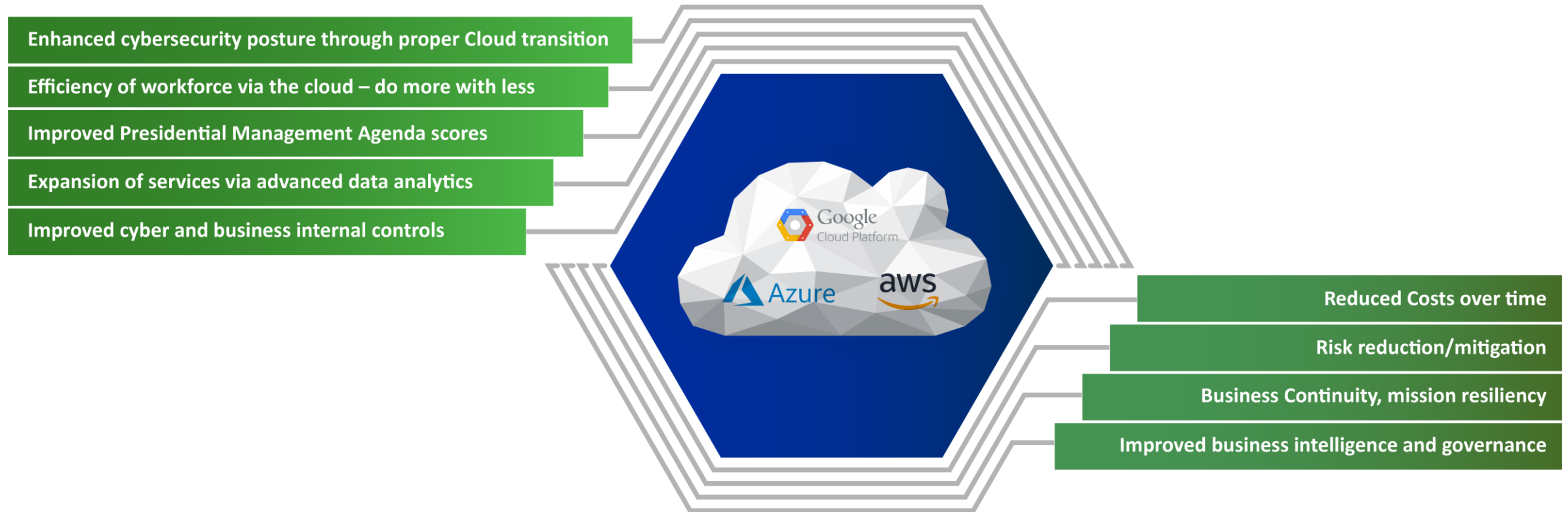
JANUARY 2021



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WHY CLOUD ?



WHY CLOUD ?

CAPABILITY- CLOUD ENABLES USE OF MODERN TECHNOLOGIES

- **Capability** is the ability to leverage all the new service delivery and decision support capabilities of modern computing including “big Data”, analytics, IoT, and AI/ML using 5G to support your mission. If your apps and information are not cloud-enabled, these capabilities will not be available.

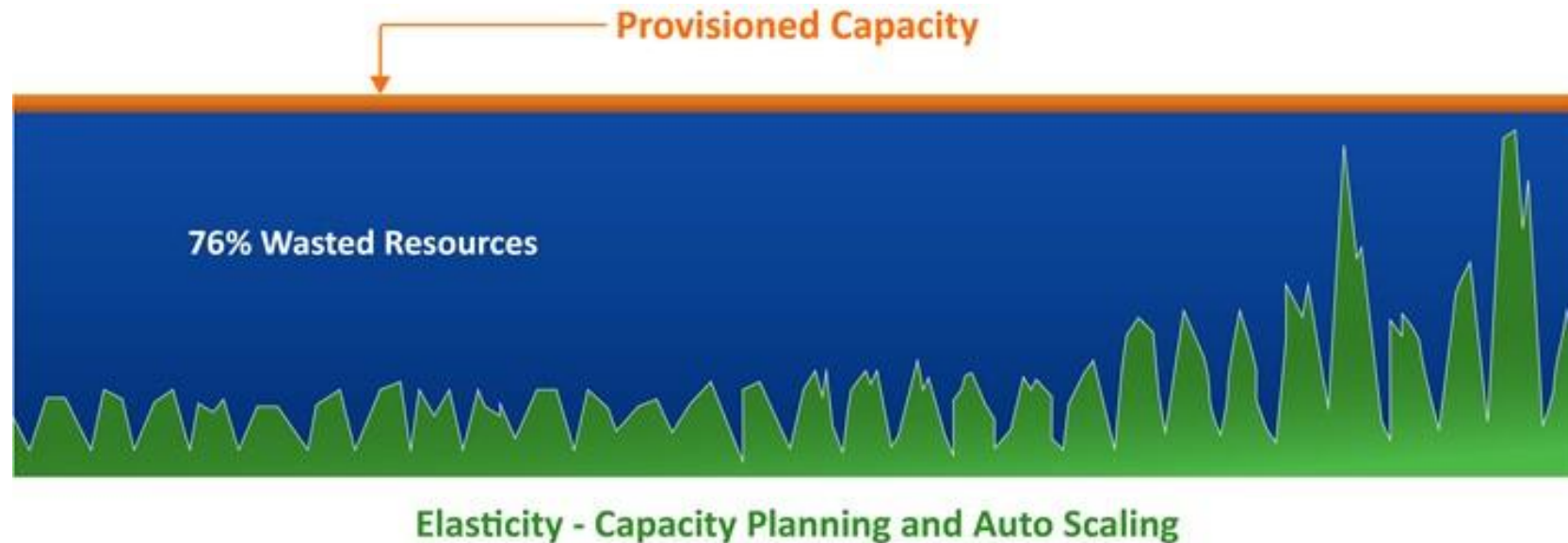


While your current technology may seem OK to you, what does it look like to the new generation of workers and customers who do everything on their mobile phone?

WHY CLOUD ?

CAPACITY - ON DEMAND RESOURCES CONSUMPTION

- **Capacity** refers to the fact the current data center architecture requires you to “guess” what your level of user demand on your systems will be and provision for the highest level. Cloud allows you the flexibility to get support at the level of your “real time” needs, in order to provide the best user experience.

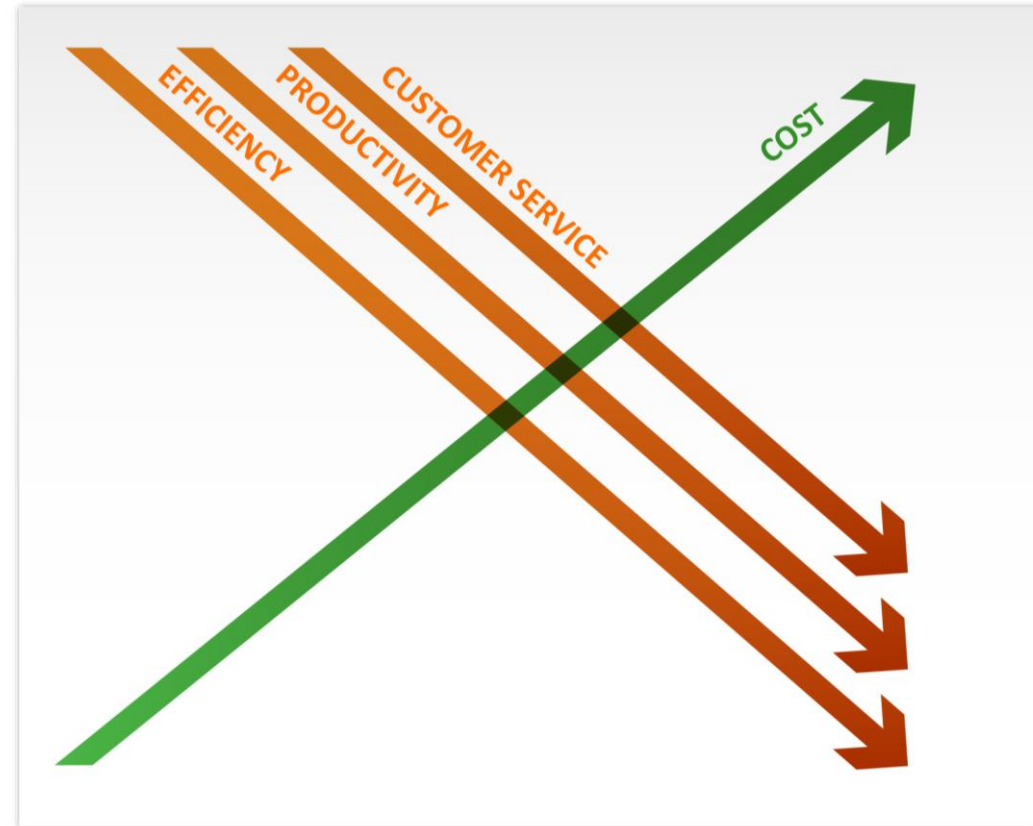


Pay for what you need, while having confidence your systems will be always be accessible for customers

WHY CLOUD ?

RISK - LEGACY HOSTING POSES INCREASING STRATEGIC RISK

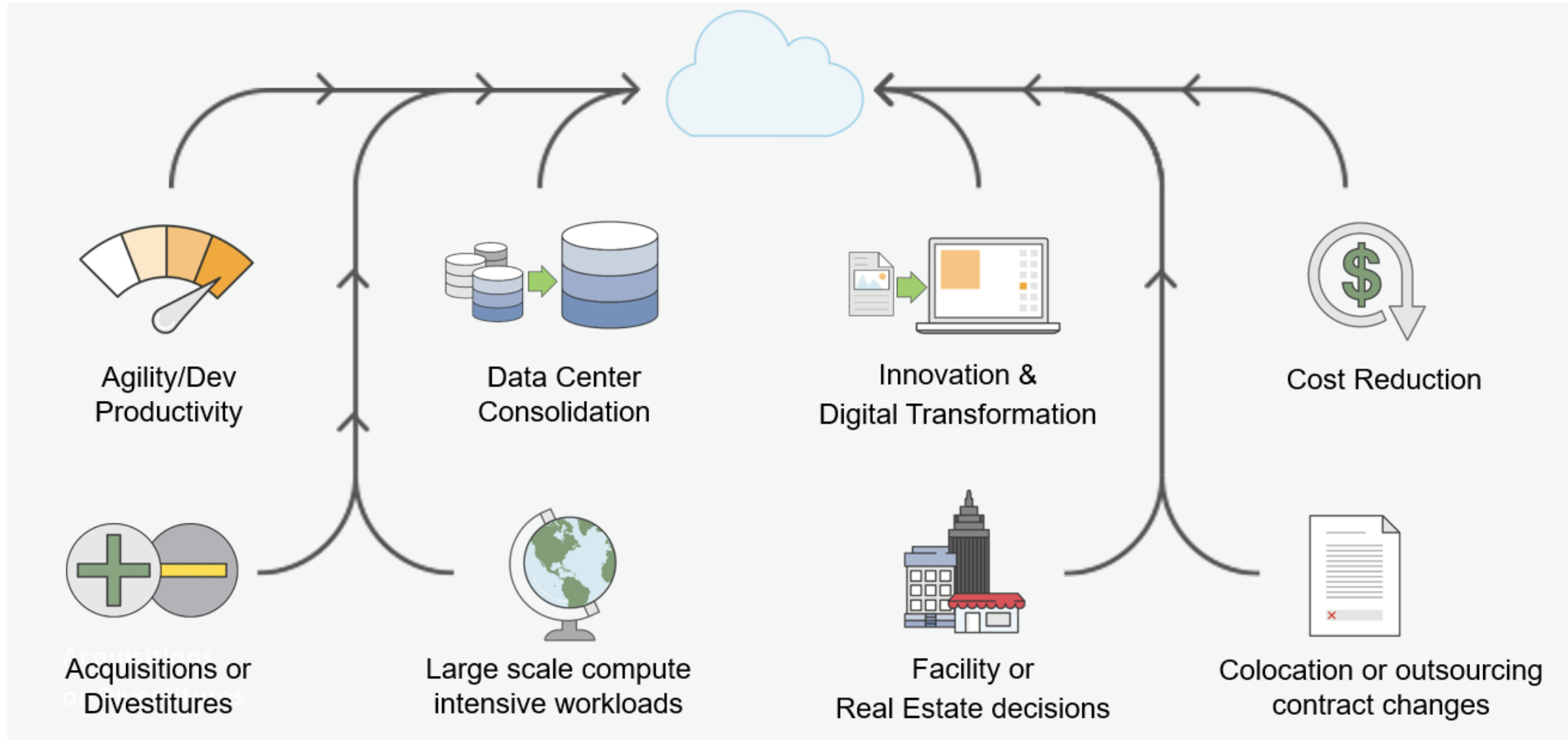
- **Risk** is that you will be stuck on old technology that no one supports anymore, and maintenance costs will continue to rise.
- Worse yet, these old systems will become security risks.
- No Access to Benefits of the cloud
- Limited Choices



In the end, you will have to make a dramatic and more costly decision, while maintaining inefficient old systems.

WHY CLOUD ?

CLOUD ADOPTION – COMMON DRIVERS SUMMARIZED



CLOUD ADOPTION SERVICE MODELS (REPRESENTATIVE)



CLOUD ADOPTION FEATURES, VALUE-ADD AND OUTCOMES

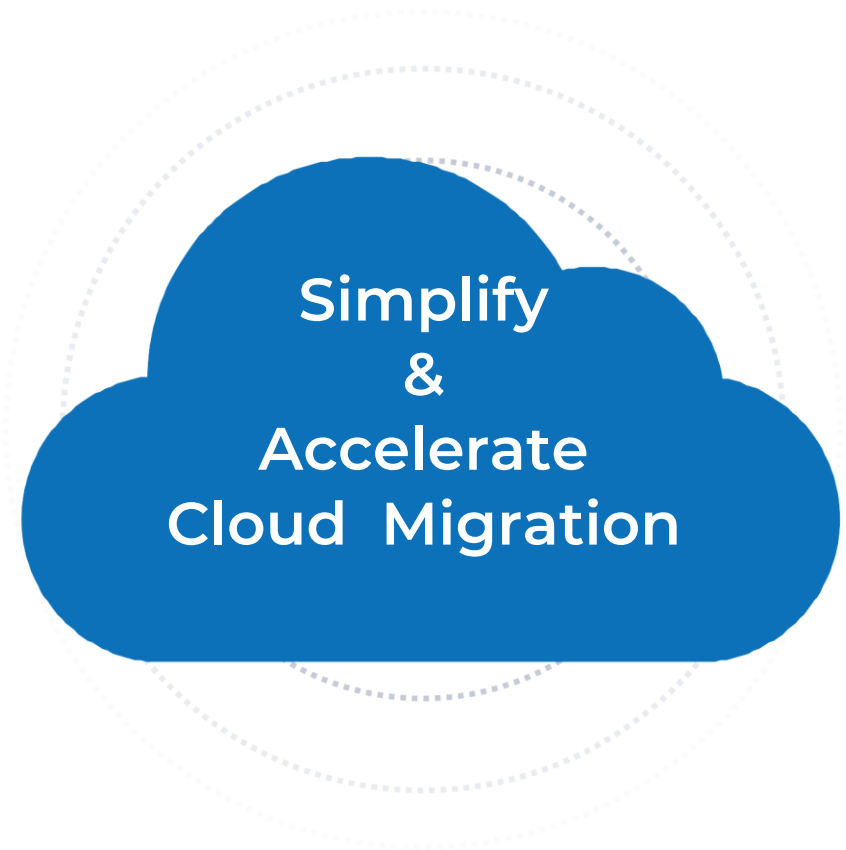
Features

Managed
Cloud Services

Turn-Key
Migration

DevSecOps
Automation

Flexible Pricing
Models

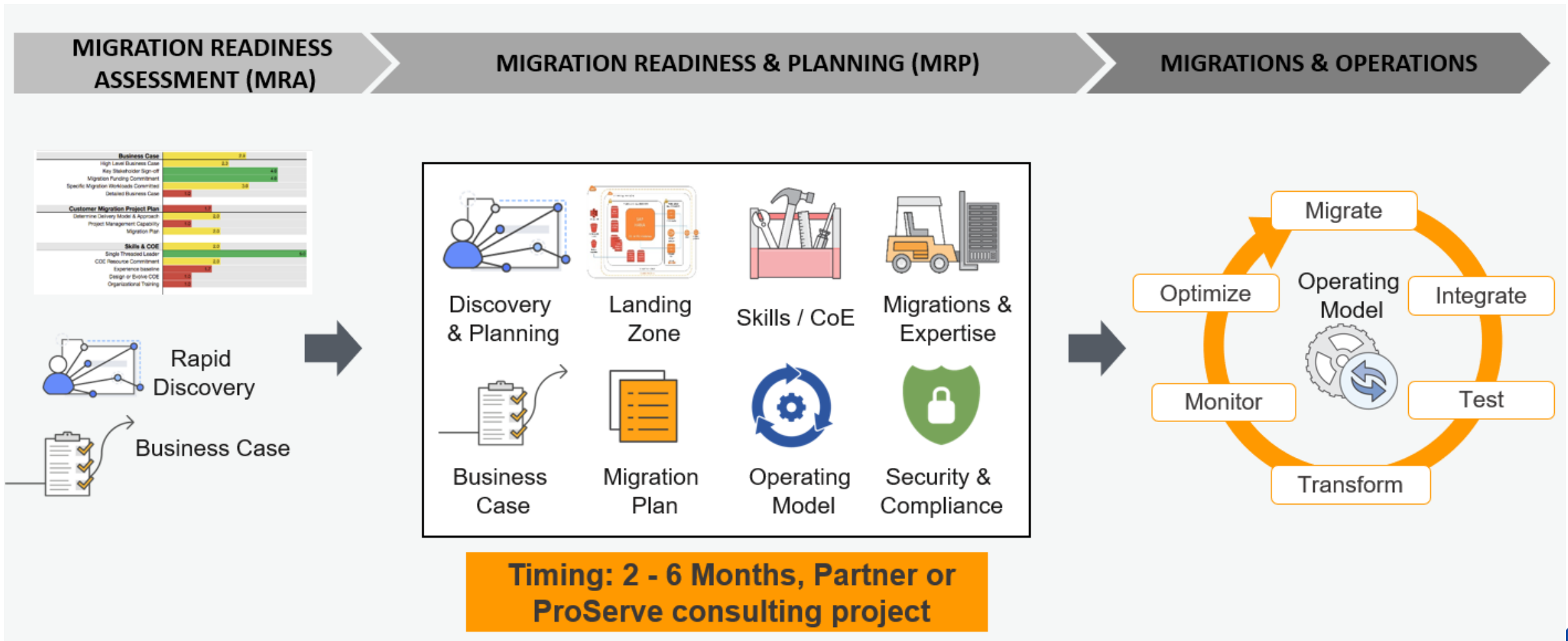


**Simplify
&
Accelerate
Cloud Migration**

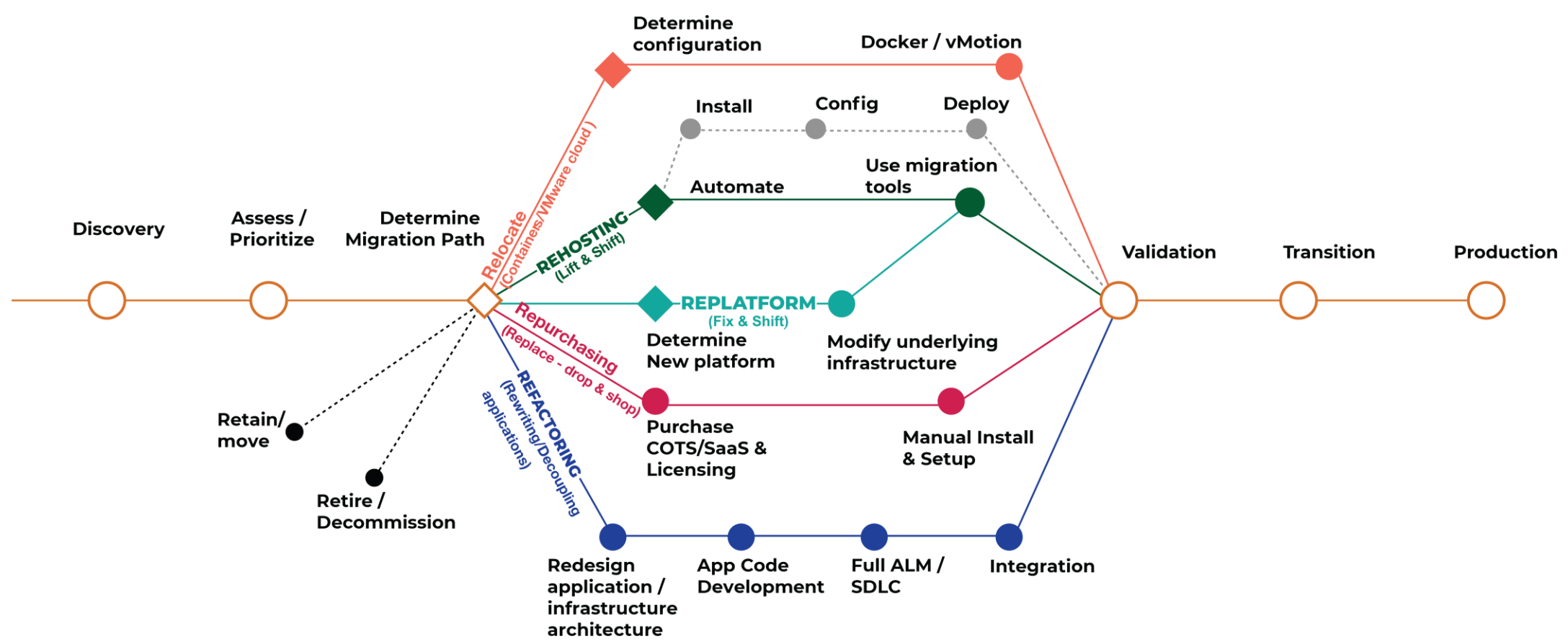
Value Add and Outcomes

- ✓ Agility and Scalability
 - ✓ Innovation & Emerging Technologies
 - ✓ FedRAMP and FISMA ATO Compliant Environment
 - ✓ Interoperability and Portability
-
- ✓ Infrastructure as Code
 - ✓ Security and Compliance (ATO) Support
 - ✓ Disaster Recovery
 - ✓ High Availability
 - ✓ CI/CD with Zero Downtime Deployments

CLOUD ADOPTION JOURNEY

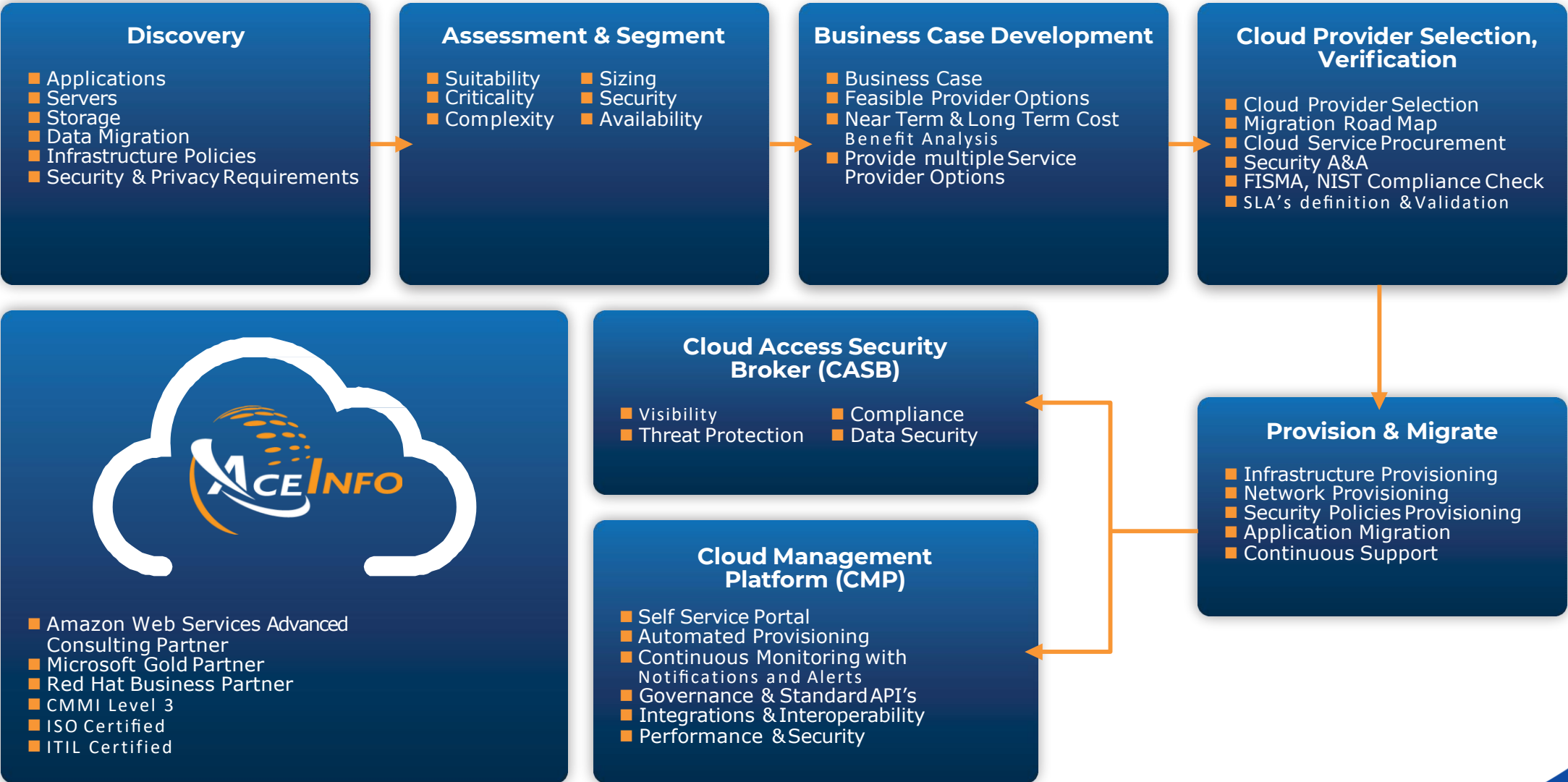


CLOUD ADOPTION STRATEGY (7 R's)



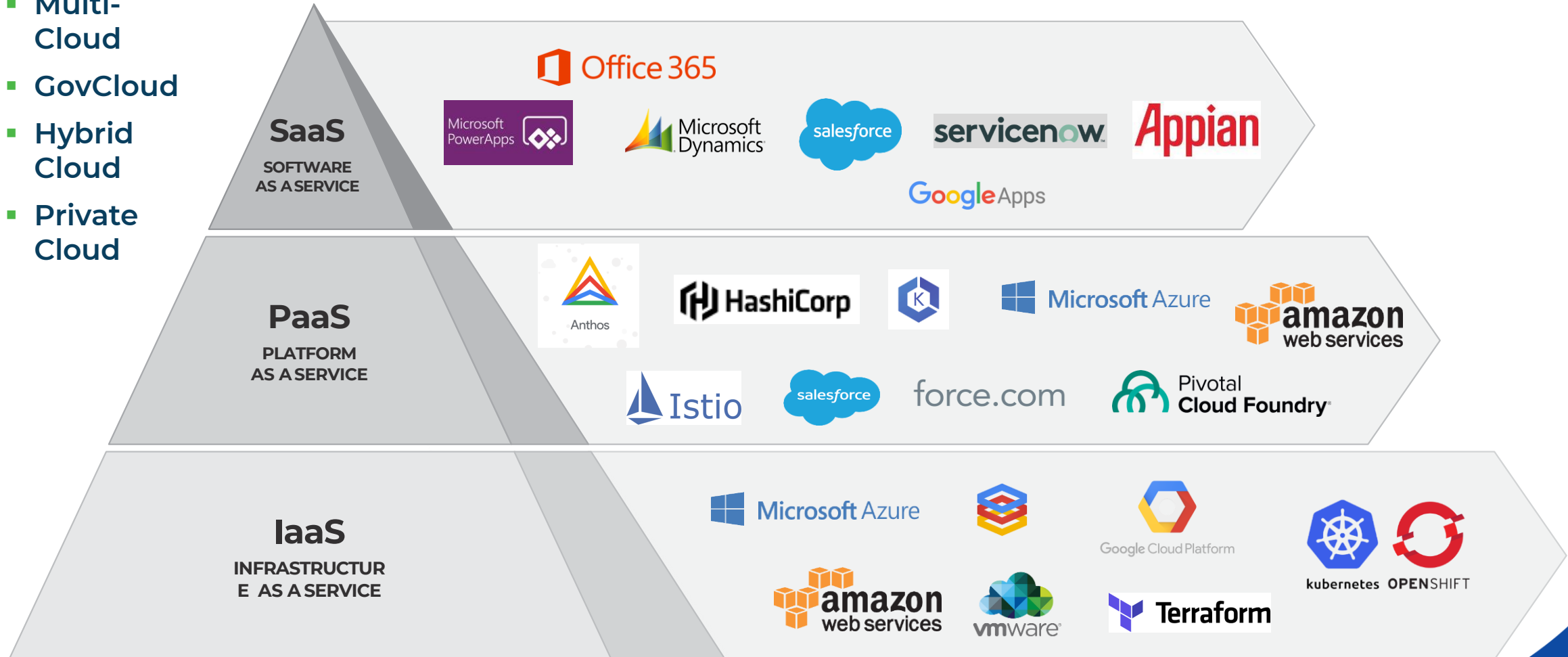
Migration Tools & Services: Storage Migration, Server Migration Services, Database Migration Service, Application Discovery Service, AWS Migration Hub, Azure Site Recovery and Cloud Endure

HIGH LEVEL CLOUD MIGRATION APPROACH



CLOUD SERVICE PROVIDERS, PLATFORMS AND TOOLS

- Multi-Cloud
- GovCloud
- Hybrid Cloud
- Private Cloud



CLOUD ADOPTION ACCELERATOR PLATFORM

<https://cloudaccelerator.doveltch.com>

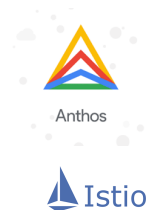
Features and Benefits

Turnkey Solutions
Self-Service Templates
Empowers Dev, Ops and Sec
Accelerate Innovation
Rapid Development & Deployment

Containers / Kubernetes Orchestration

OpenShift, AWS EKS, GKE, AKS

DHS FEMA : 350 - 500
Containers
U.S. Air Force : 100+
Containers
USDA : 100+
Containers
Census : 100+
Containers
USCG : 50+ Containers
NIH : 50+ Containers
ITG/CoE Lab : 30 Containers



Istio, Service Mesh and Google
Anthos

**Cloud
ACCELERATOR**
Powered by Dovel Technologies

Cloud Adoption and Digital Modernization Accelerator
Accelerate Innovation | Simplify and Optimize Cloud Journey
Concept Overview | Cloud Adoption Playbook

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Infrastructure as Code

- Create Cloud Infrastructure
- View Cloud Infrastructure
- Nexus/Docker Hub
- Terraform Templates
- Cloudformation Templates
- Ansible Templates
- Resources +

Application Development

- Create MicroService and Application
- Running Applications
- Github Repository
- Quality Assurance PAL
- Discover Dovel Technologies
- Application Prototypes +
- Resources +

DevSecOps

- Code Security Scan
- View Code Scan Results
- View All Pipelines
- OpenShift
- API Management
- Regulatory Compliance
- Resources +

Intelligent Services / Analytics
(Powered by AWS and GCP Services)

- Apollo - Intelligent Data Processing (IDP)
- Apollo - View IDP Jobs

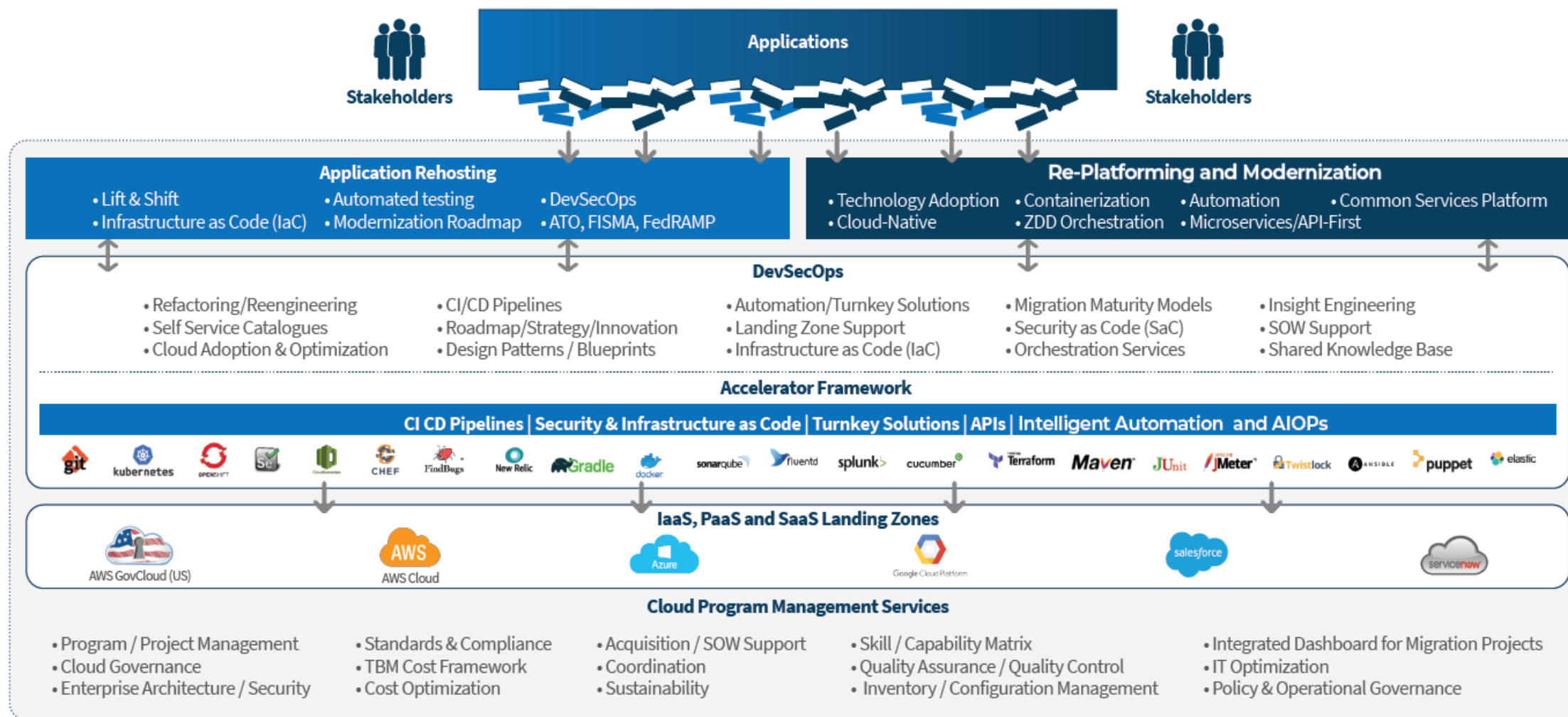
Monitoring and Logging

- Hygieia
- Splunk
- New Relic
- ELK Stack
- Slack
- Sonarqube
- Resources +

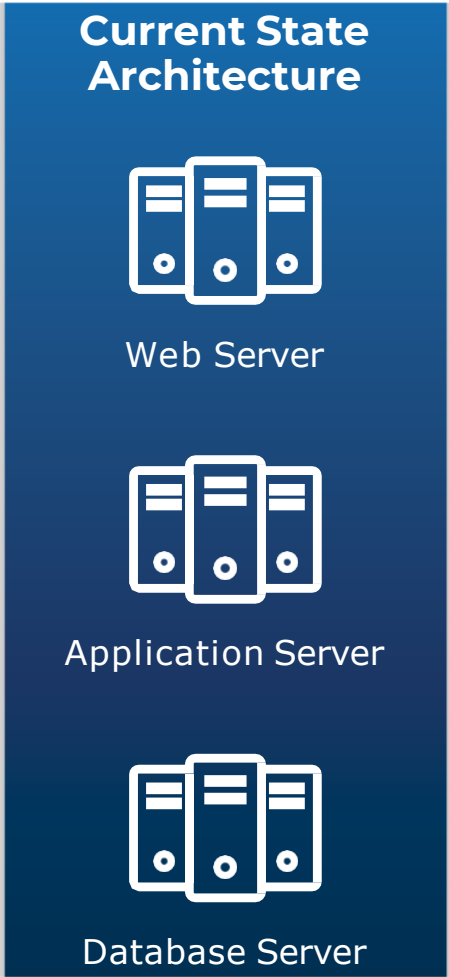
High Performance and Quantum Computing

- AWS Cloud HPC
- Google Cloud HPC

CLOUD ADOPTION ACCELERATOR CONOPS

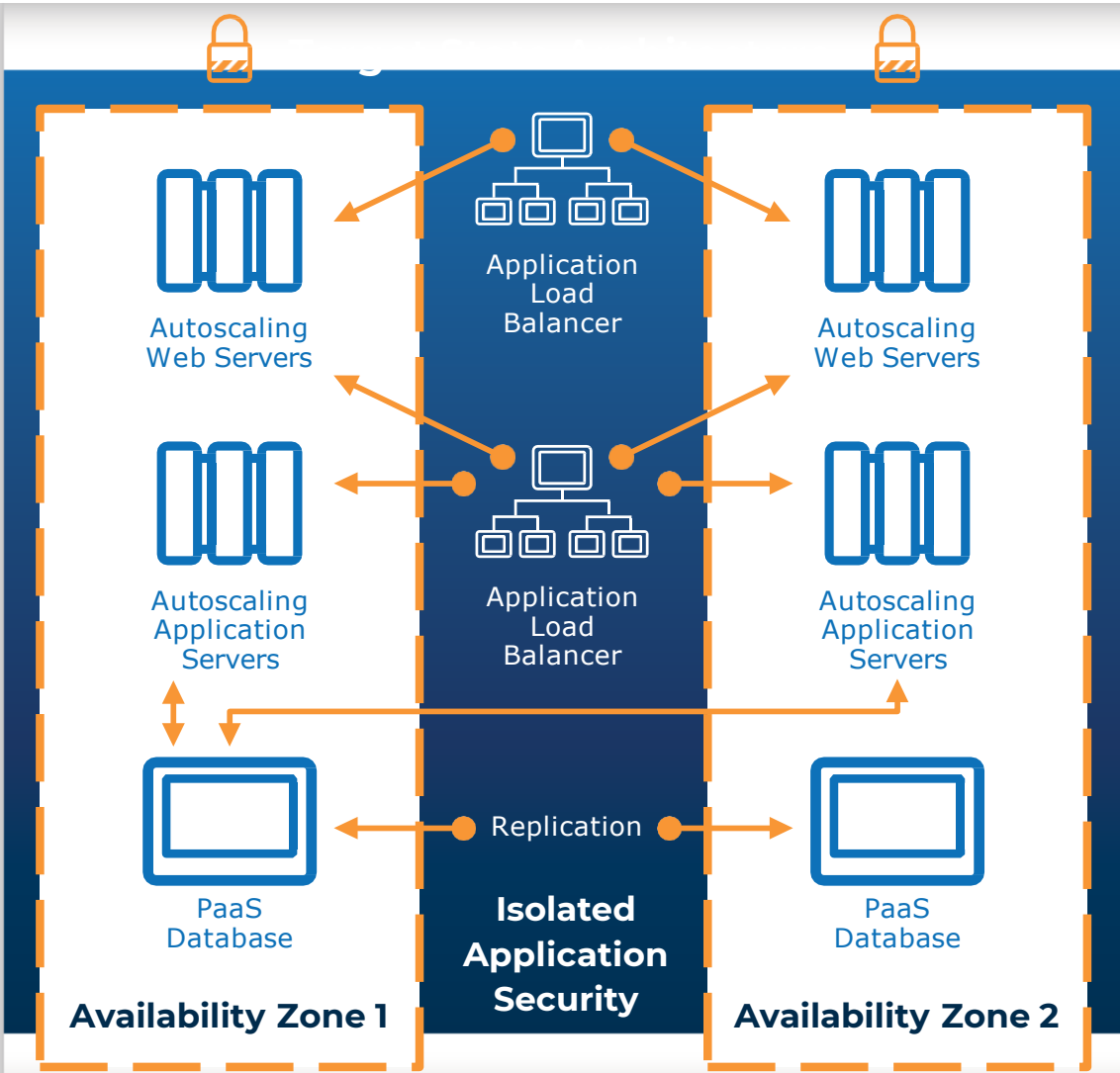


CLOUD PLATFORM TARGET ARCHITECTURE (MULTI AND HYBRID CLOUDS)



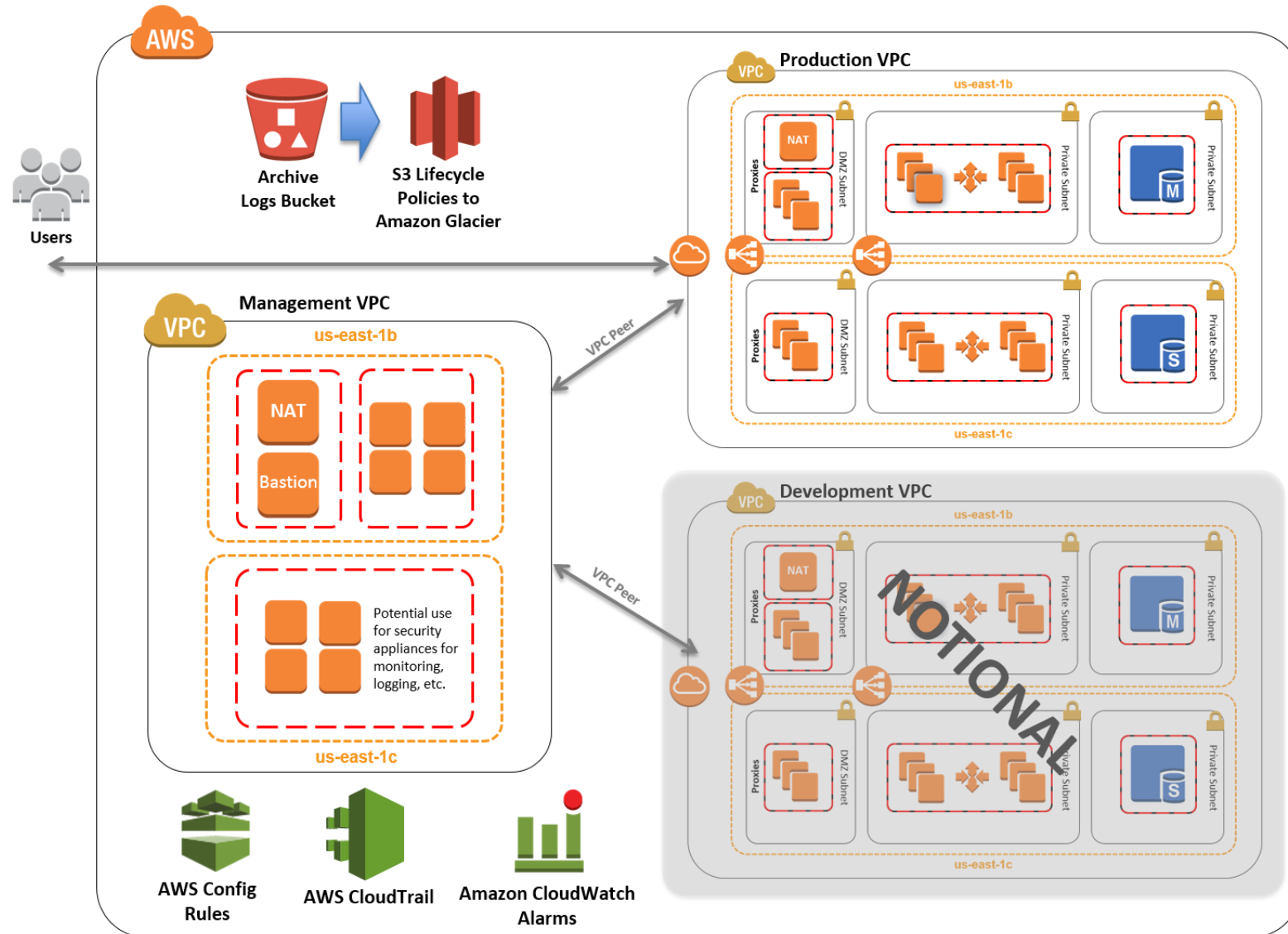
Target State Features:

- Autoscaling
- Elastic
- Highly Available
- Built-in DR
- Isolated Security
- PaaS Database



STANDARDIZED ARCHITECTURE FOR NIST 800-53 FISMA LEVEL COMPLIANCE

BAKED-IN TEMPLATES FOR AUTHORITY TO OPERATE (ATO) ACCELERATION

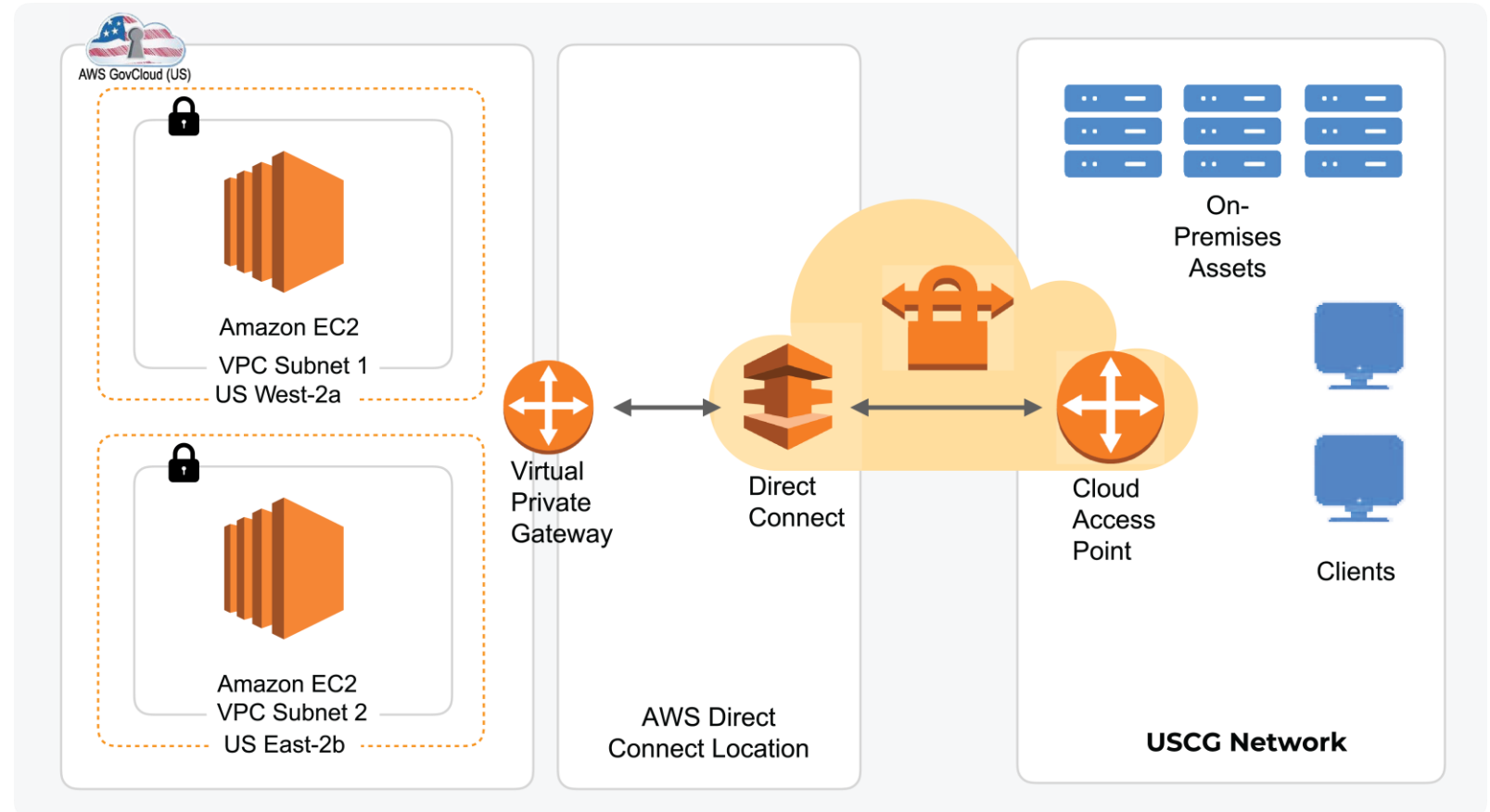


HYBRID CLOUD ARCHITECTURE

AWS Global Infrastructure

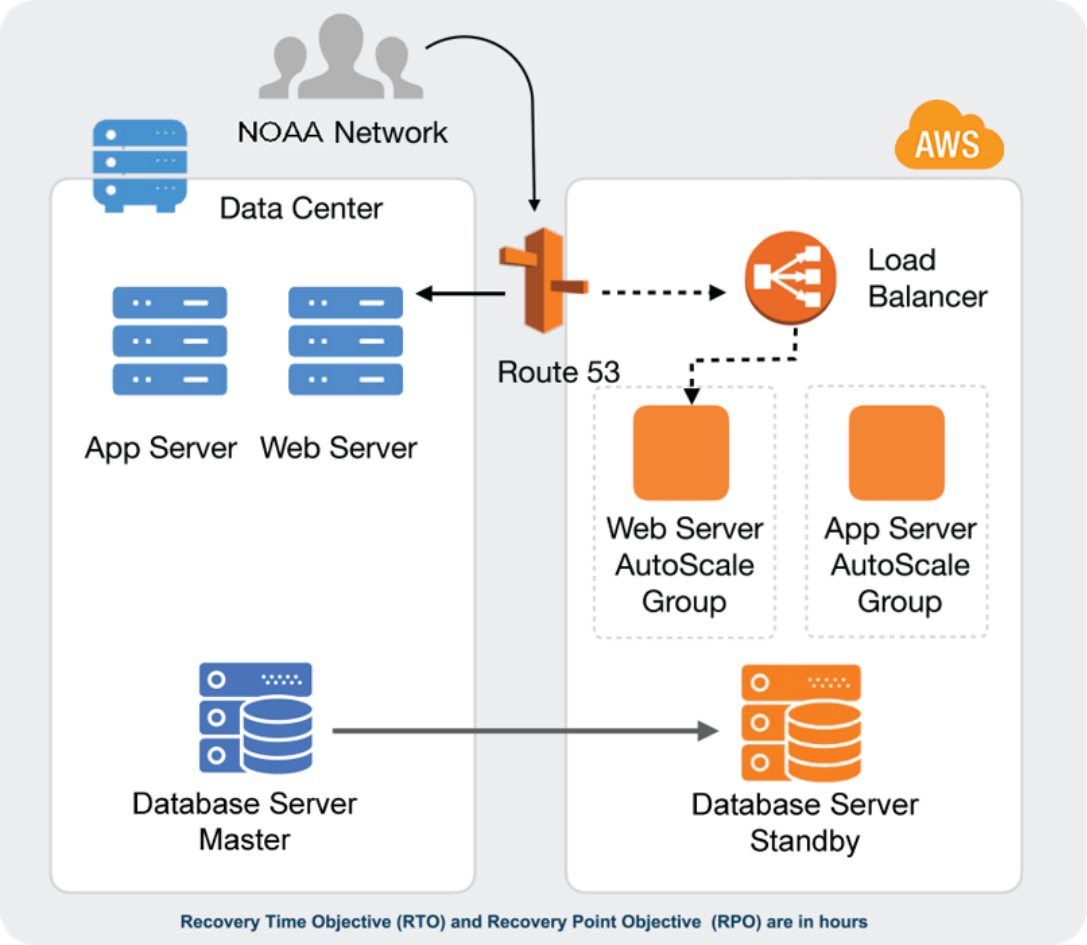
- 2 US GovCloud Regions (US East and West) with 6 Availability Zones
- 21 Geographic Regions with 66 Availability Zones.

- **Infrastructure as Code**
- **Security as Code**
- **Automation as Code**

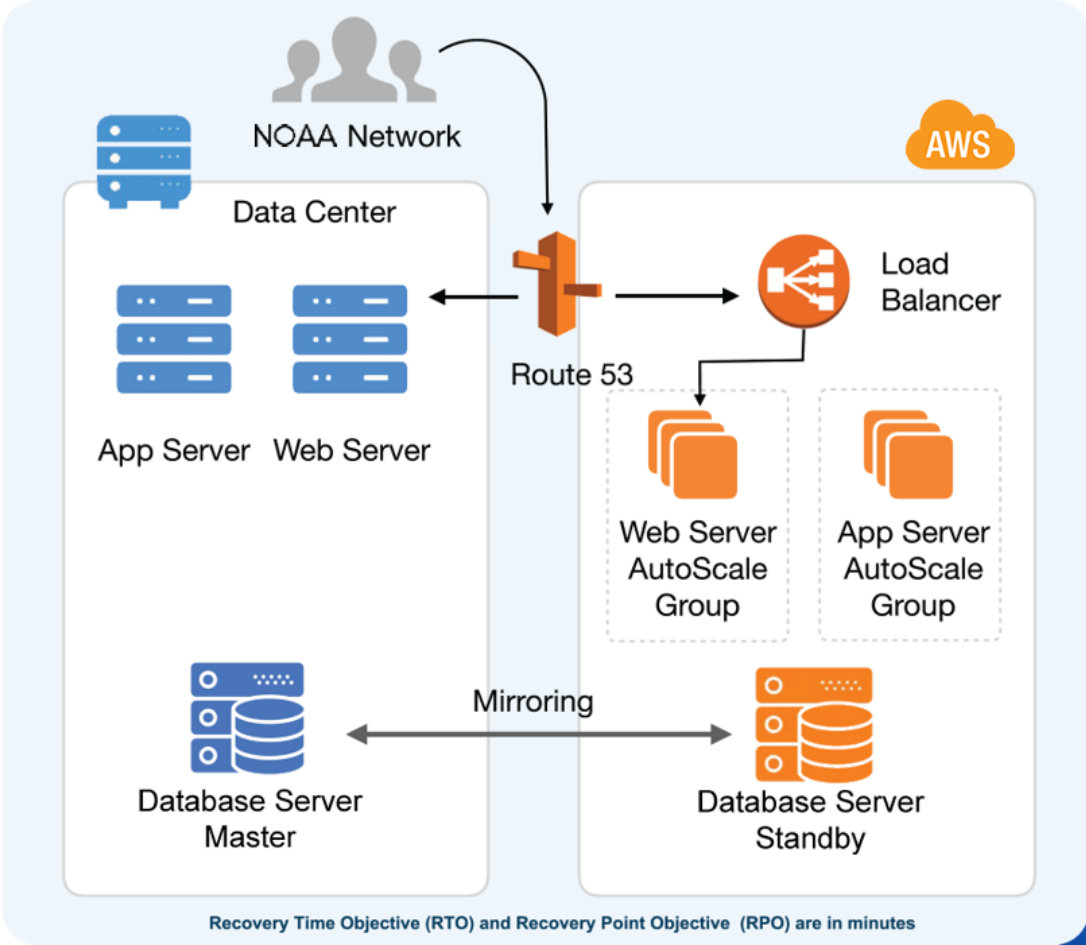


DISASTER RECOVERY AND FAILOVER ORCHESTRATION

Warm Standby Active



Active -



AGILE SOFTWARE DEVELOPMENT AND DEVSECOPS

- **15+ years of Agile Software Development**

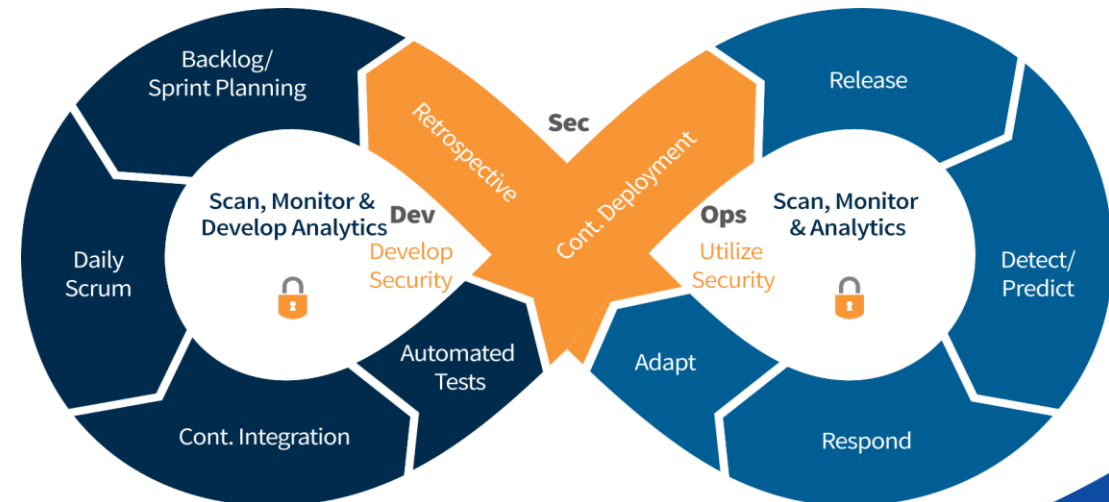
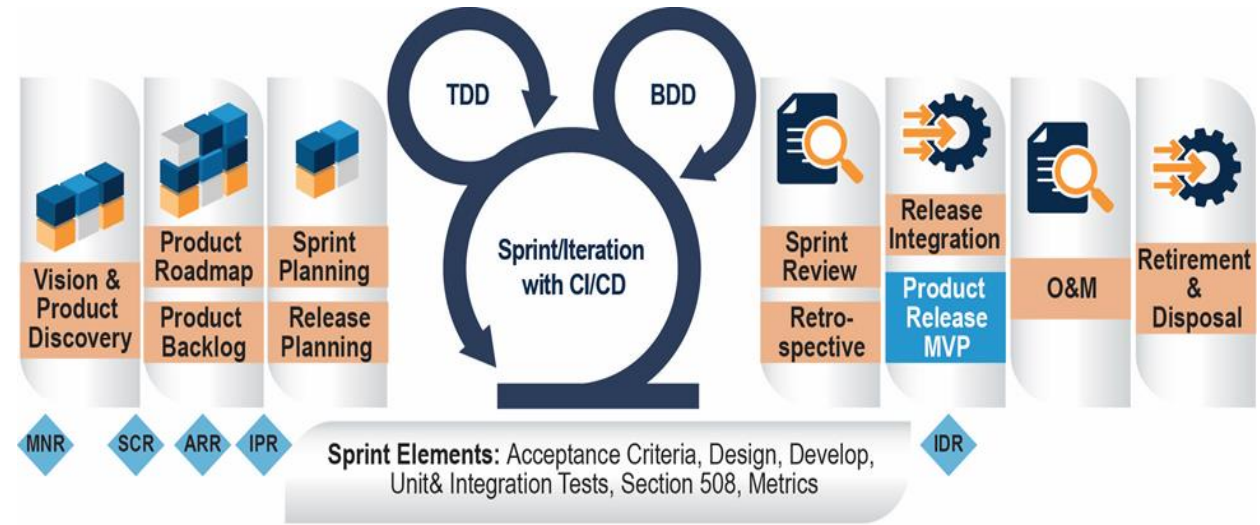
- Scrum, Kanban, Lean, SAFe
- Design Thinking
- User Centric Design (UX/UI)
- Responsive Web Design
- APIs, Microservices
- Opensource, COTS and GOTS
- Developed and maintain 500+ applications

- **350+ Agile Experienced and Certified Resources**

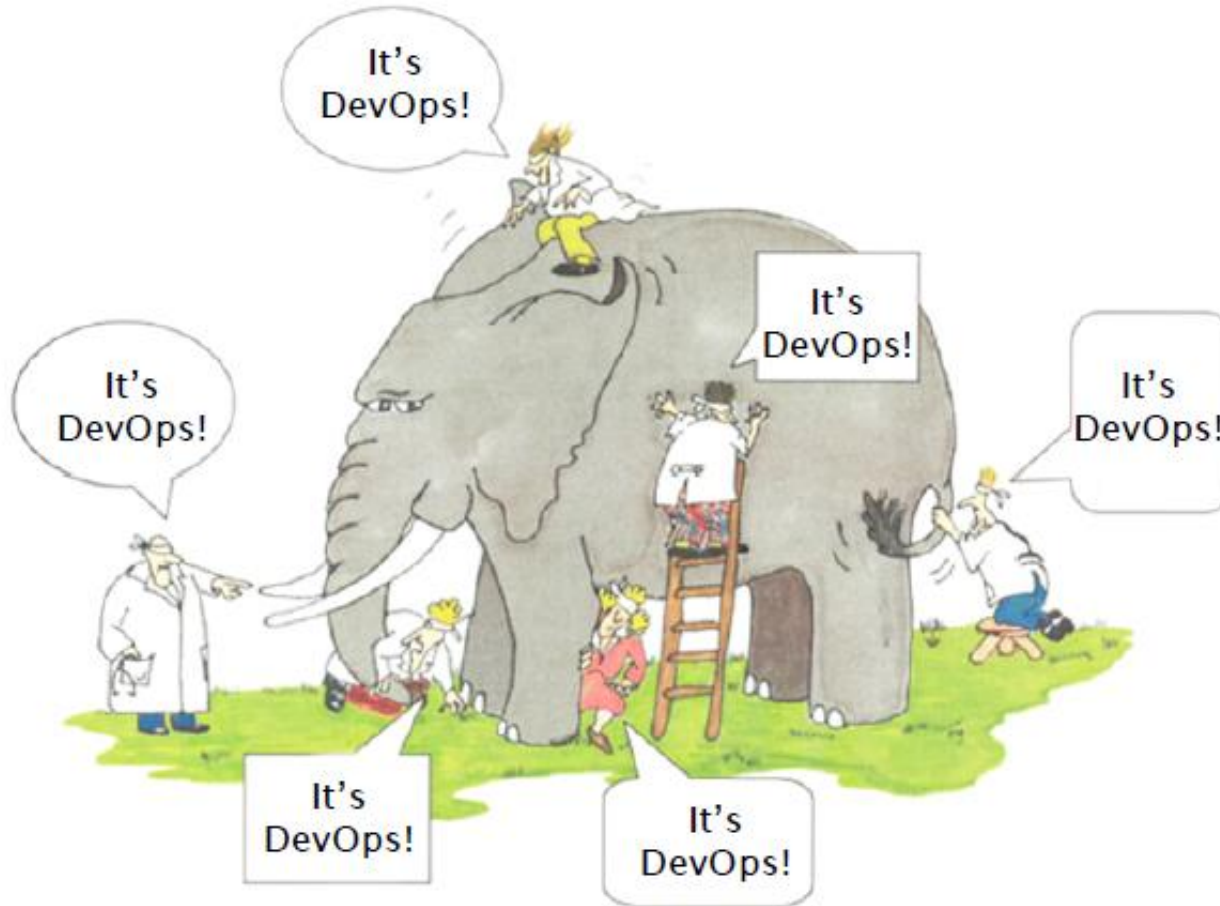
- Certified Scrum Masters/Professionals
- Agile Project Managers
- PMI-ACP and SAFe Certified Practitioners

- **5+ Years of DevOps / DevSecOps Implementation**

- 30+ DevOps/DevSecOps Engineers
- 100+ Redhat Engineers (50+ Certified)
- 100+ Cloud Engineers (25+ Certified)



What is DevOps?



Let's try again - What is DevOps for customers ?

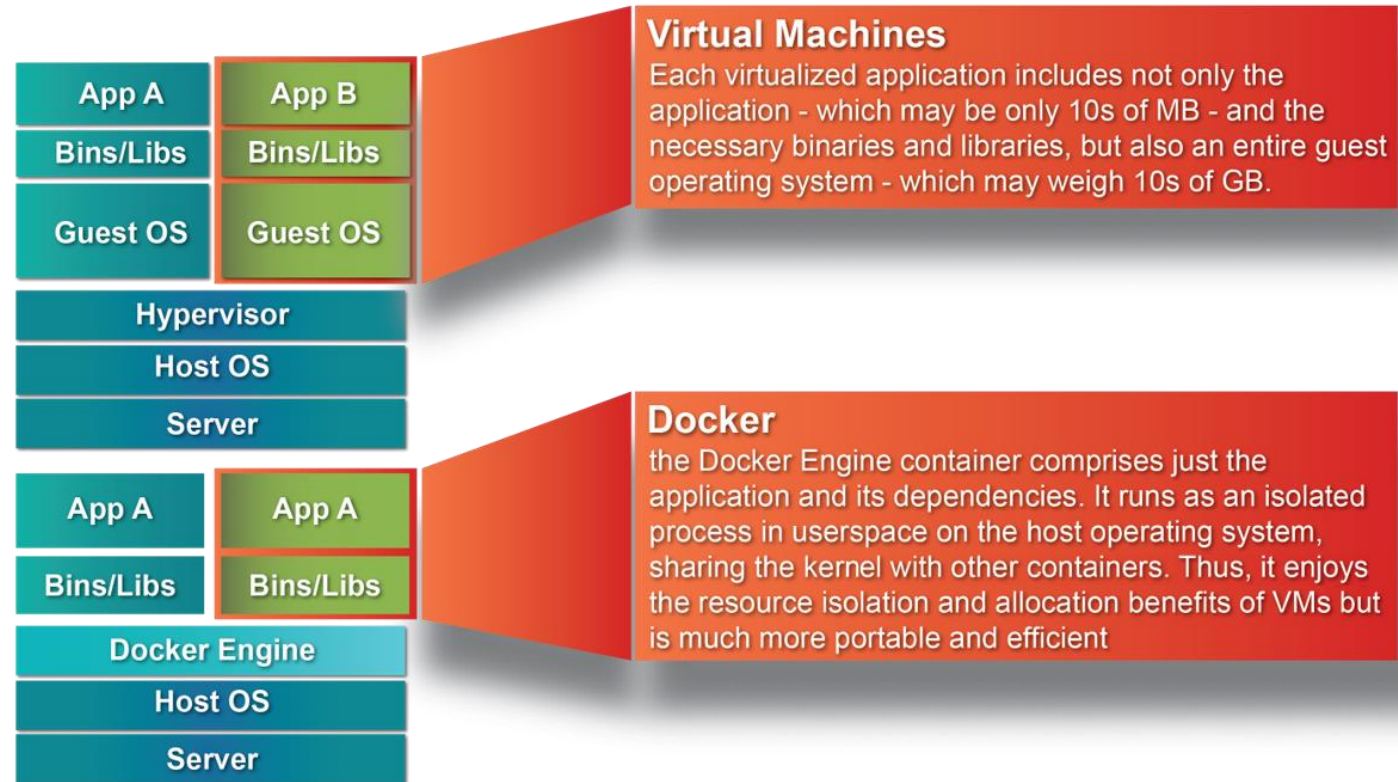
- ✓ Collaborative Culture
- ✓ Automation as Code
- ✓ Effective & Efficient Toolchain
- ✓ Monolithic to Microservices
- ✓ Containerization and ZDD
- ✓ High Availability in Cloud
- ✓ AIOps (Predict and Prevent)

Never send a human to do a machines job



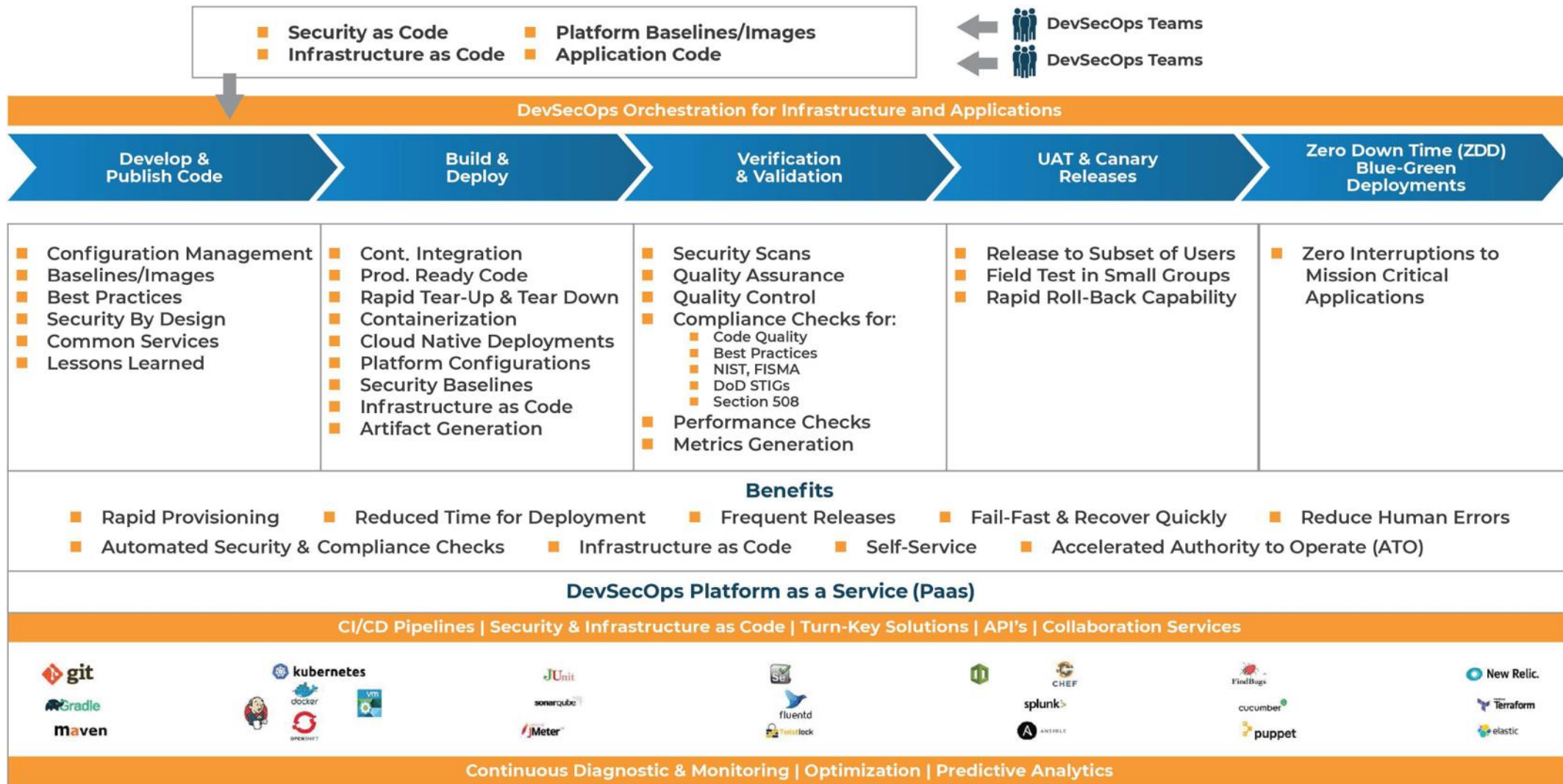
CONTAINERIZE APPLICATION BUILDS AND SOFTWARE LIBRARIES

- Package Once and Deploy Anywhere
- Immutable and Reusable
- Standardized Environments
- Effective Configuration Management
- Portability and Integrity
- Cloud Readiness
- Interoperability
- Infrastructure as Code
- Security as Code / ATO
- Rapid Provisioning and Scaling



DEVSECOPS ORCHESTRATION FOR INFRASTRUCTURE & APPLICATIONS (CI/CD)

AGGREGATED ARTIFICATS FOR AUTHORITY TO OPERATE (ATO) ACCELERATION



KUBERNETES BASED HIGH AVAILABILITY AND SELF-HEALING

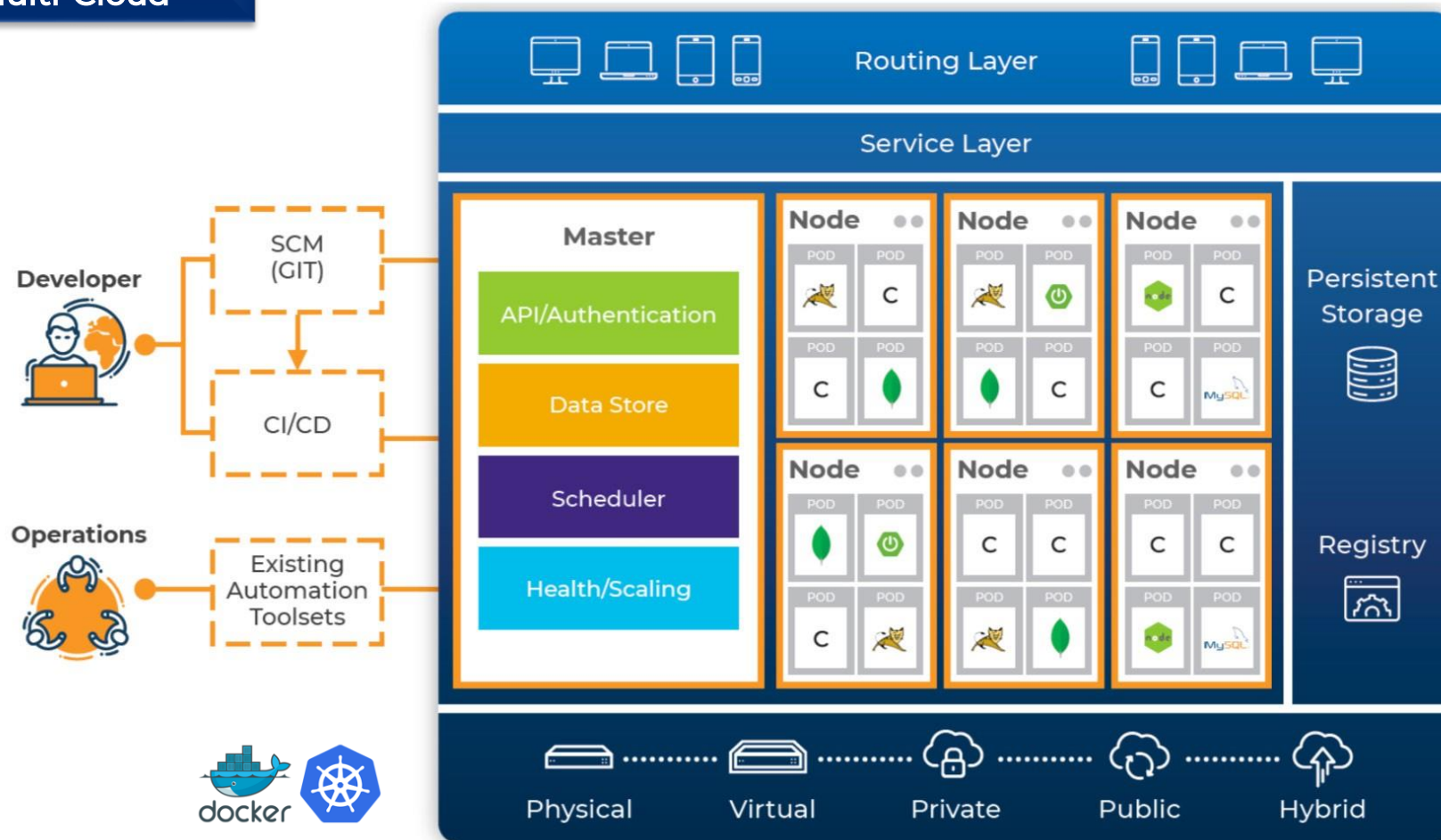
On Premise, Hybrid and Multi-Cloud

■ Turnkey Solutions

- Service Catalog / Templates
- Rapid Provisioning/Scaling
- Security and ATO
- Zero Downtime Deployments (ZDD)
- Self-Healing
- Monitoring and Logging
- AIOPs

■ Turnkey Platforms

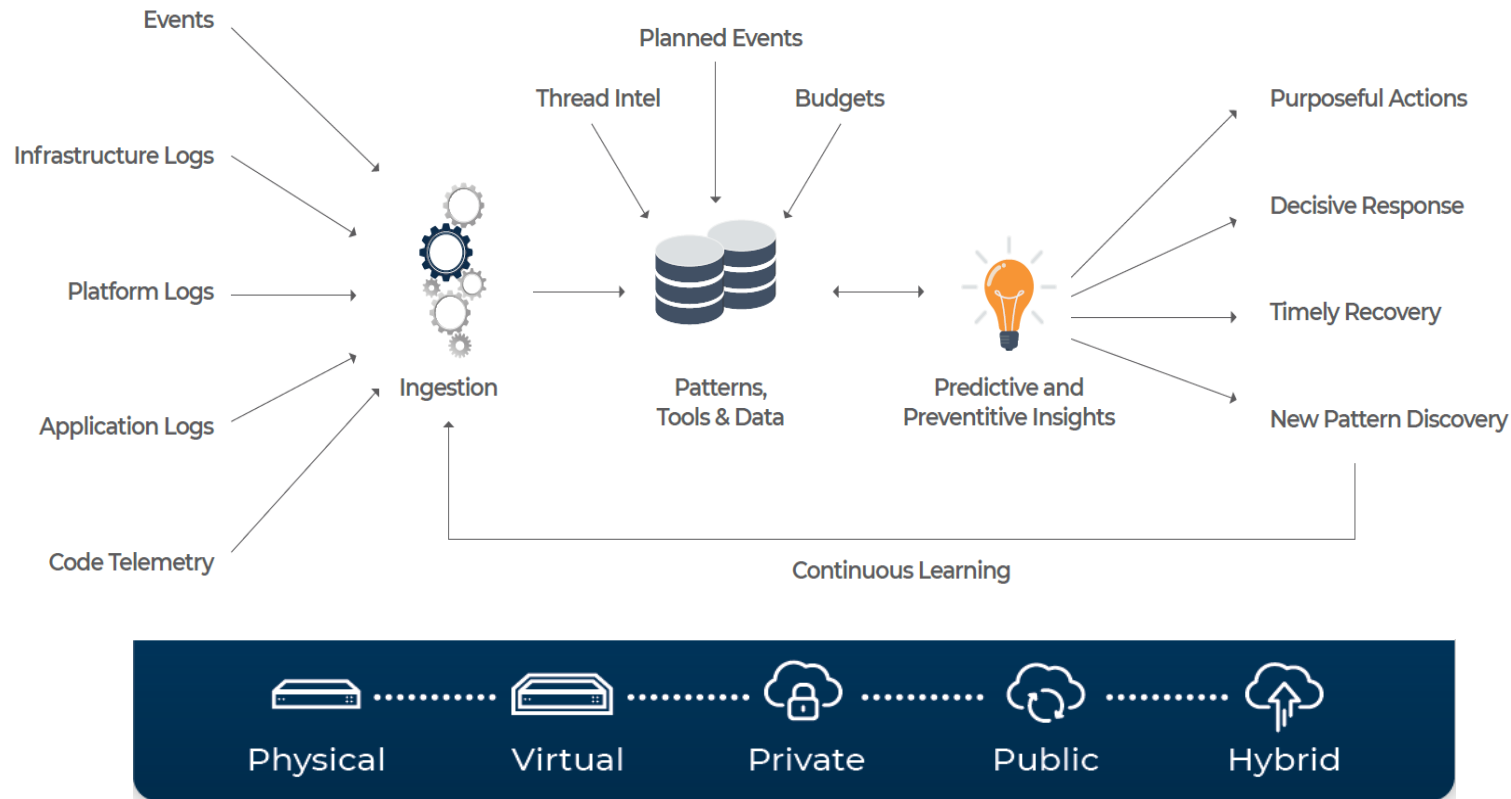
- Kubernetes
- Openshift
- Redhat Openshift
- Google Cloud (GKE)
- AWS (ECS, EKS)
- AZURE (CS, KS)



ARTIFICIAL INTELLIGENCE FOR IT OPERATIONS (AIOPS)

PREDICTIVE INTELLIGENCE AND PREVENTIVE MEASURES

Full Stack Observability for Distributed Architectures

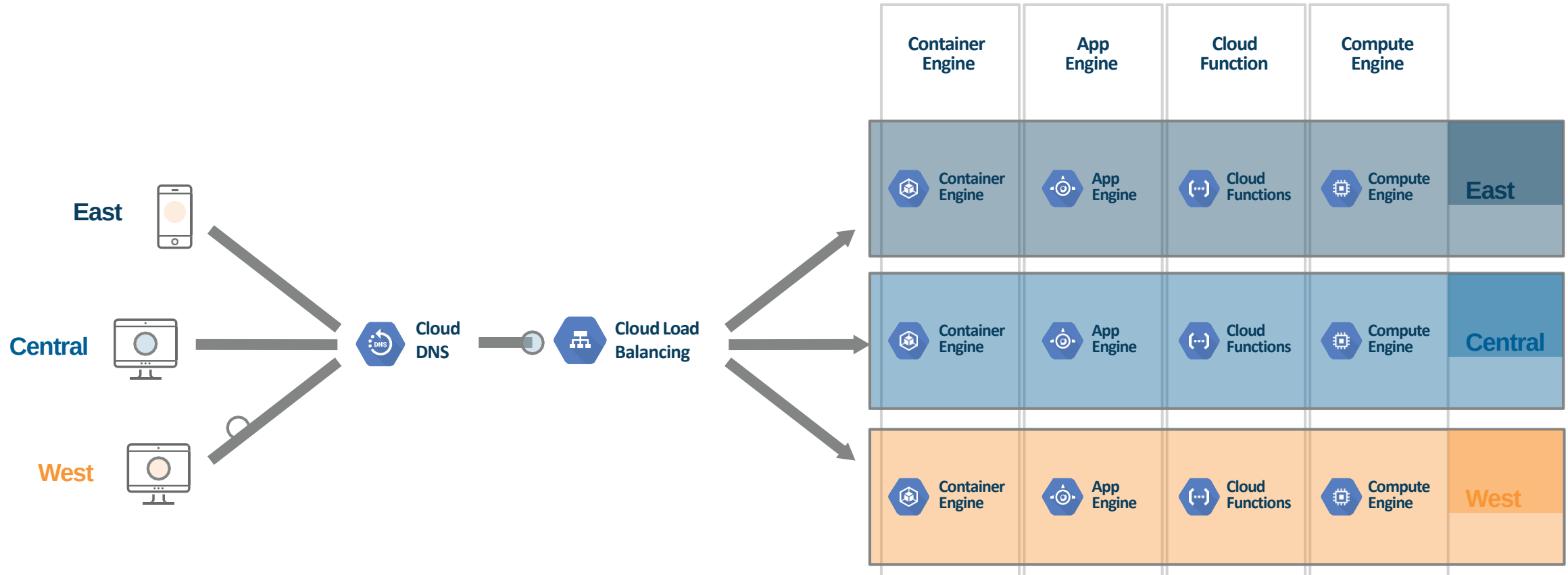


Use Data Lake and AI / ML

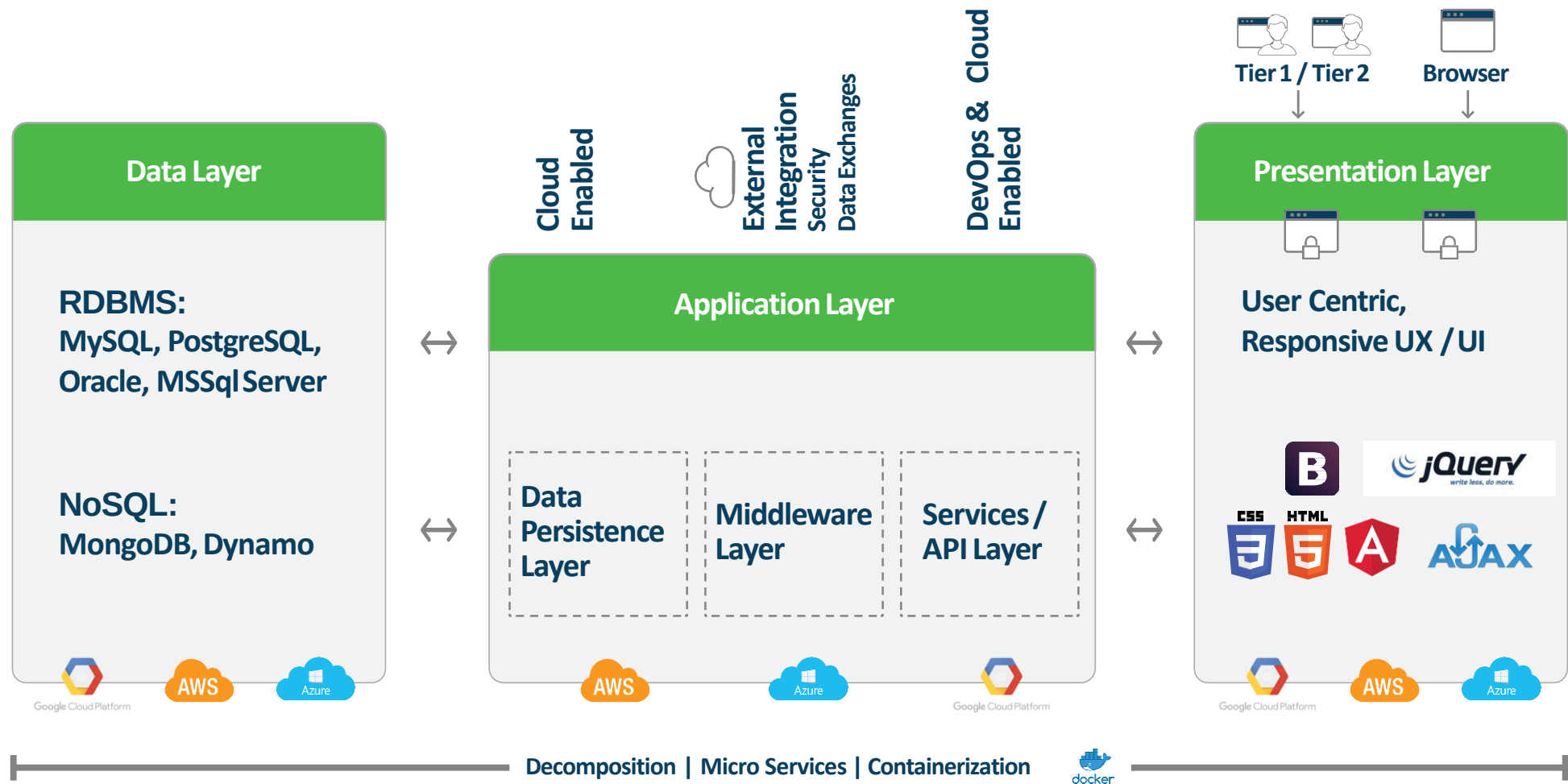
- ✓ Ingest & Process Data (3 Vs)
- ✓ Log Analytics and Insights
- ✓ Event Correlation
- ✓ Pattern Recognition
- ✓ Speed Root Cause Analysis
- ✓ Lower Noise | Detect Anomalies
- ✓ Automate Remediation Proactively
- ✓ Unify IT Metrics and Business KPIs
- ✓ Improve Customer Experience
- ✓ Reduce Cost | Improve Efficiency

KUBERNETES CONTAINER ORCHESTRATION (GOOGLE CLOUD)

- Multi Region Failover & High Availability



CLOUD NATIVE ARCHITECTURE FOR APPLICATION MODERNIZATION



CLOUD CAPABILITIES AND EXPERIENCE

 Advanced Consulting Partner Public Sector Partner	25+ Cloud Implementations Private, Public and Hybrid Clouds	 Gold Data Platform Gold Application Development Gold Cloud Platform	
	8+ Years of Cloud Computing AWS, MS Azure, Google Cloud, Salesforce and Red Hat Consulting Partner		 Red Hat Business Partner
	5000+ Servers, 200+ Applications and Petabytes storage in various Clouds		
	100% Corporate Websites and Applications in AWS/Azure/Google Clouds		
	100+ Cloud Skilled Engineers, 50+ AWS/Azure/Google Certified Architects		
			

DHS and DoD (TSA, USCG, FEMA, and Airforce)	Department of Commerce (Census and NOAA)	USDA, EPA, and DOJ COPS	Health and Human Services (NIH, SAMHSA and ACF)
- DOD STIG Compliant Hybrid/Private Cloud, hosting 2500+ Servers, 75+ Applications - Self-service provisioning with Single Pane of Glass for Cloud Management - DevSecOps: Docker, Kubernetes, Openshift Container Orchestration - Air Force CON-IT and FEMA PIVOT Systems Modernized and implemented in GOGO cloud	1500+ Servers, 200+ Applications, Dev/Test Environments, 100+ Databases and petabytes of storage - Data Lakes and Analytics - Private, Public, & Hybrid Cloud - Managed Services for AWS and Azure commercial and GovCloud - DevSecOps: Anisible and Openshift Jenkins CICD	- Migrated 40+ applications of DOJ COPS to Azure Cloud. - Modernized 35+ legacy monolith applications to cloud native architecture - Several .NET, open-source, and GIS based applications migrated to AWS and Azure Cloud - DevSecOps, Jenkins CI/CD, Containers, Microservices - Private, Public, & Hybrid Cloud, Salesforce, and ServiceNow	- Modernized legacy Grants Management System for SAMHSA into AWS GovCloud - Migrated legacy COMET application to AWS (<60 days) with zero service interruption - Modernized and maintained Grants and Healthcare applications and infrastructure in AWS Cloud - Reduced TCO of existing systems by 40%

CLOUD ADOPTION AND DIGITAL MODERNIZATION EXPERIENCE

AceInfo's Agile Experience	USDA EAS				USDA	USDA	NOAA	US EPA	DOJ	DOC	DHS	HHS
	DHS FEMA PIVOT	AirForce CON-IT	GSA	Other	FAS	FSA	NWS	OPP & OEI	COPS	US Census	US Coast Guard	SAMSHA
Number of Staff	60	60	33	80+	50+	60	50	25	40	124	140	45
Number of Agile Teams	6	6-7	4-5	30+	5	5	4	4	3	Varies	Varies	5
Agile SELC Methodologies	Scrum	Scrum	Scrum	Scrum, Kanban	Scrum	Scrum	Scrum, Kanban	Scrum, Kanban	Scrum	Hybrid	Hybrid	Scrum
DevOps/DevSecOps	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Portfolio Size	5 Applications	1 System	1 System	86 Applications	40+ Applications	140+ Applications	20 Applications	60+ Applications	50+ Applications	98 Applications	60 Applications	5 Systems
Degree of Complexity / Size	300+ User Stories 78 Sprints/year	300+ User Stories 120 Sprints/year	180+ User Stories 72 Sprints/year	3000+ User Stories 120 Sprints/year	1000+ User Stories 78 Sprints/year	1,800+ User Stories 120 Sprints/year	500+ User Stories 24 Sprints/year	800+ User Stories 27 Sprints/year	700+ User Stories 36 Sprints/year	Varies	Varies	1000+ User Stories 120 Sprints/year
Project Cloud Attributes												
SaaS		✓		✓				✓		✓		
PaaS	✓			✓				✓	✓	✓	✓	
IaaS	✓			✓				✓	✓	✓	✓	
COTS		✓	✓	✓	✓	✓		✓	✓	✓	✓	✓
Mobile			✓	✓					✓			
Custom	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓
DevOps	✓	✓		✓		✓	✓	✓	✓	✓	✓	✓
Support	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Public/Private Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NOAA Cloud/ AWS	N/A	Azure GovCloud	AWS Cloud/ GovCloud	Hybrid: Private Cloud/AWS/Azure	AWS GovCloud
Number of Users	Nationwide	5,800	2,500	N/A	3,000	6,000	3 Million	1,300	50,000	20,000	150,000	20,000

INFRASTRUCTURE CAPABILITIES AND EXPERIENCE

- **Managed Infrastructure (IT) Services**

- Data Center Support
 - 10,000 Servers, Petabytes of Data, 99.99% SLAs
- Security and Compliance (FISMA, DOD STIG)
- Disaster Recovery and Failover (High Availability)
- IT Automation and Optimization
- Virtualization and Private / Hybrid Cloud
- Network Engineering (LAN, WAN)
- Operating Systems (Linux, Windows)
- Software Technology Stack (COTS, Opensource)
- Storage Management
- Enterprise Middleware and Integration
- Databases and Elastic Clusters
- MS SharePoint and Office365 Services

- **Enterprise IT Customers / Contracts**

- US Census Bureau (ISSA) – 130 FTEs
- US Coast Guard (Core Technology Services) -150+ FTEs
- NOAA National Weather Service (WCCIS) – 100+ FTEs
- NIH Data Center Support – 50+ FTEs
- Environment Protection Agency – 20+ FTEs
- Department of Justice COPS – 20+ FTEs
- USDA DISC including Security Services (ATO)
- USDA Security Operations Center (ASOC)

- **Engineering Staff and Partnerships**

- 500+ Systems/ Network Engineers and Administrators
- Technology Partnerships and Certifications



CLOUD ADOPTION SERVICE MODELS WITH ESTIMATED PRICING

SINGLE APPLICATION MIGRATION

Service Description

- Strategy and Planning
- **Re-host (“Lift and shift”)** of Application including its database and integrations from on-premise to cloud
- **Re-platform** technology stack of on-premise application and migrate to cloud (Complexity: Moderate/High)

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of Databases and Data Volume
- Number of Application Integrations (External & Internal), Web Pages, Userbase
- High Availability and Disaster Recovery requirements
- Re-hosting vs. Re-Platforming the application

Estimated Cost Per Application Per Annum (Labor Only) based on Complexity*

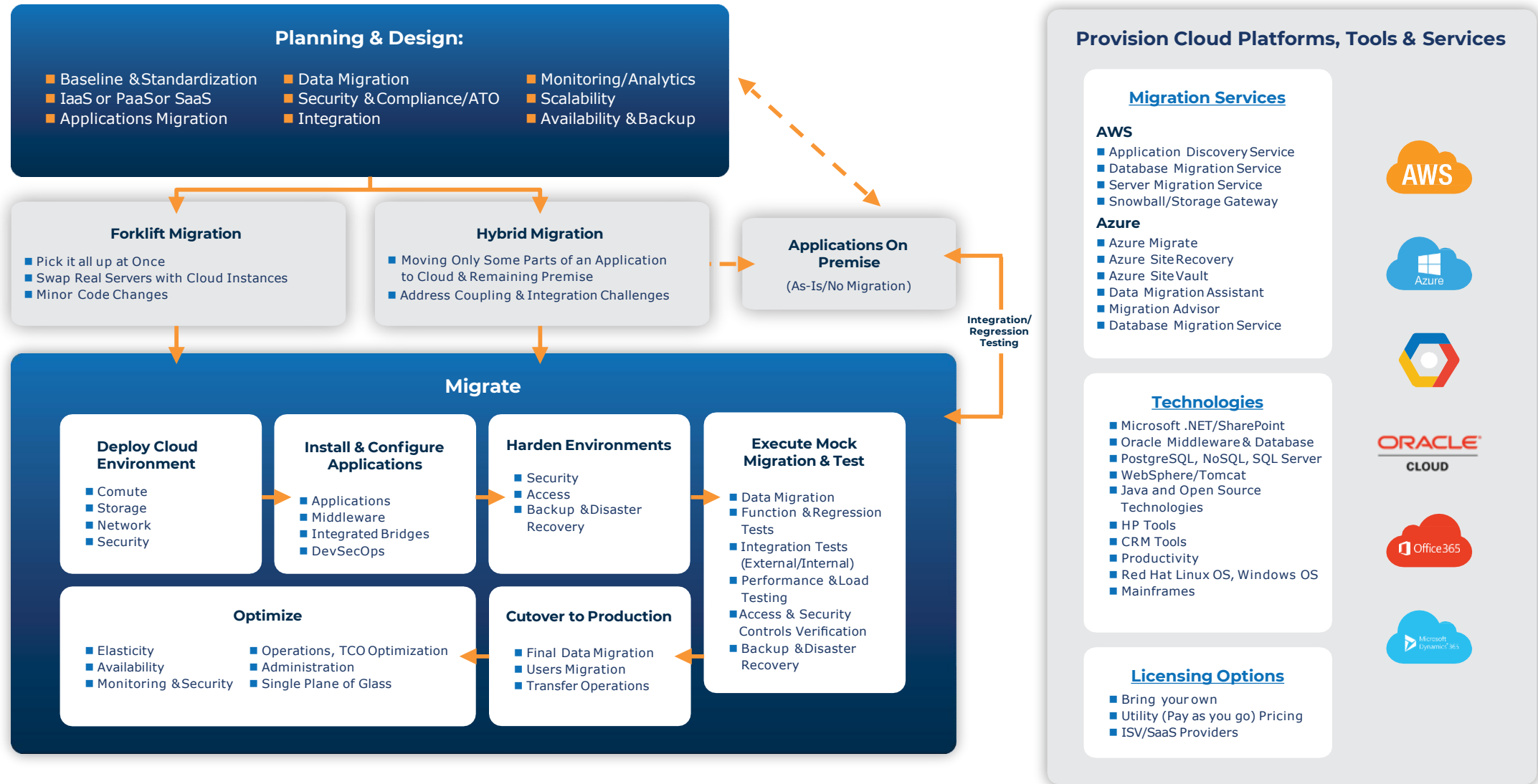
- Low: \$200K - \$400K
- Moderate: \$400K - \$800K
- High: \$800K and Above

Deliverables

- Production ready running App in Cloud (Public/GovCloud) including ATO
- Basic CI/CD Pipeline, High Availability and Disaster Recovery

*Cloud Infrastructure costs not included

RE-HOSTING (LIFT AND SHIFT) IMPLEMENTATION



Service Description

- Strategy Consulting and Planning: Assist in conducting Discovery, Assessment and Recommend the Cloud Migration Path
- **Modernize** and **re-engineer** application using modern day and cloud enabled Architecture with APIs, Microservices and containers
- Deploy the modernized application in Cloud

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of Databases and Data Volume
- Number of Application Integrations (External & Internal), Web Pages, Userbase
- High Availability and Disaster Recovery requirements
- Number of legacy technologies and frameworks being used

Estimated Cost Per Application (Labor Only) based on Complexity*

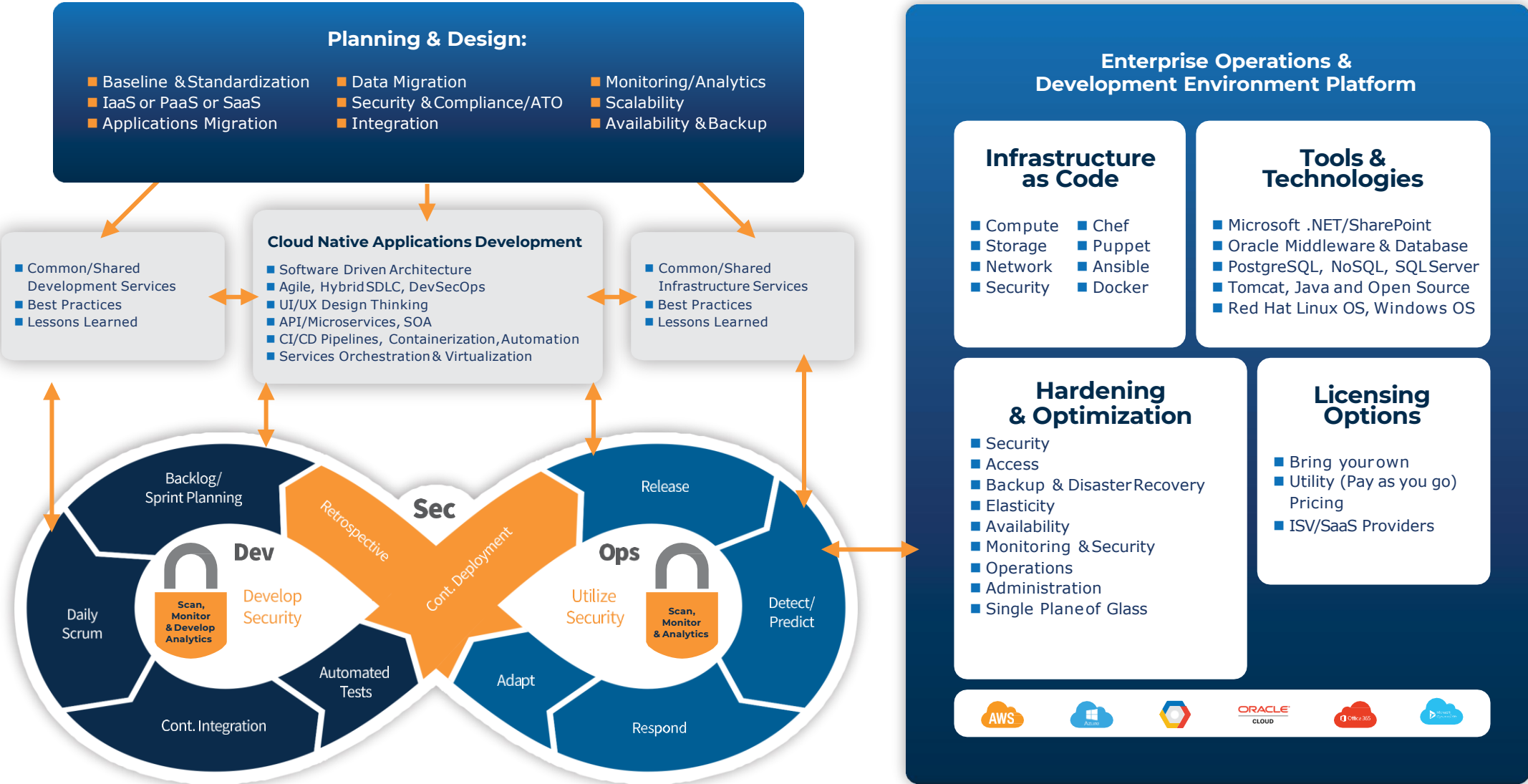
- Low: \$500K - \$800K
- Moderate: \$800K - \$2M
- High: \$2M and Above

Deliverables

- Production ready running Application in Cloud (Public/GovCloud) with **cloud native architecture** including **ATO**
- CI/CD Pipeline, High Availability and Disaster Recovery

*Cloud Infrastructure costs not included

RE-PLATFORM AND APPLICATION MODERNIZATION



FULL APPLICATION PORTFOLIO MIGRATION



Service Description

- Strategy Consulting & Planning including Discovery, Assessment/Prioritization and Determine Migration Path
- Application Migration (Lift & Shift), Re-Platform and Application Modernization
- **Deliver efficiencies** w.r.t Infrastructure as Code, Security as Code, Re-Usable processes and templates

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of Applications in the portfolio, requiring Re-hosting, Re-Platforming and Modernize
- Number of Databases and Data Volume, Application Integrations (External & Internal), Web Pages, Userbase
- High Availability and Disaster Recovery requirements

Estimated Cost Per Application (Labor Only) based on Complexity*

- | Re-host / Re-Platform | Modernize |
|-----------------------------|---------------------------|
| ■ Low: \$200K - \$400K | ■ Low: \$500K - \$800K |
| ■ Moderate: \$400K - \$800K | ■ Moderate: \$800K - \$2M |
| ■ High: \$800K and Above | ■ High: \$2M and Above |

Deliverables

- Production ready Applications in Cloud (Public/GovCloud) with cloud native architecture including ATO
- **Single GSS** with applicable **ATO** for the entire portfolio
- **CI/CD Pipeline**, High Availability and Disaster Recovery

*Cloud Infrastructure costs not included

APPLICATIONS MAINTENANCE & SUPPORT IN CLOUD



Service Description

- Infrastructure and Platform support for applications in cloud including ATO, patching and monitoring
- Does not include application development or end user support

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of server nodes for application and databases
- Number of integrations (external/internal)

Estimated Cost Per Application Per Annum (Labor Only) based on Complexity*

- Low: \$200K - \$300K
- Moderate: \$300K - \$500K
- High: \$500K and Above

Deliverables

- System Availability and Scalability
- Systems Administration including Backup & Recovery
- Continued ATO with applicable CDM (continuous diagnostics and monitoring)

*Cloud Infrastructure costs not included

CUSTOMER VALUE PROPOSITION

- **Promote collaboration and innovation across the organization for technical excellence, customer success and business growth**
- **Incubate and operationalize solutions using emerging architectures and technologies and build intellectual capital for differentiative capabilities and customer success**
- **Simplify and accelerate Digital Modernization and complex Systems Integration through software driven architecture, reusable frameworks and turnkey solutions**
- **Leverage Cloud Computing, Automation and Intelligent Platforms to build solutions that learn, evolve, optimize and self-heal for increased efficiency and resilience**
- **Lead partnerships with industry, academia and research institutes to bring new ideas and experiment with them in our Emerging Technologies Lab for release**